Futures Curve Intelligence Engine

A complete explainer — every term, what it means, how the engine uses it, and whether the methods are actually profitable

Markets: Gold (COMEX) · BTC · ETH (Deribit) | Built from the Multi-Market Futures Curve Intelligence Blueprint
Status: Phase-1 research engine (blueprint §29 "Minimum Viable Quant Engine"), generalised to three markets
Nature: research / decision-support. **Not investment advice.**

Contents

0 · What this is & the central question

A · Market concepts
A1 Spot & futures
A2 Expiry & convergence
A3 Long, short, margin, mark-to-market
A4 Settlement: physical vs cash
A5 Basis
A6 Cost of carry (r, s, y, T)
A7 Contango & backwardation
A8 Calendar spread
A9 Term structure / futures curve
A10 Level, slope, curvature
A11 Roll yield / roll-down
A12 Convenience yield
A13 Perpetual swap & funding rate
A14 Perp basis & quarterly basis

B · Statistics concepts
B1 Rolling window & point-in-time
B2 Look-ahead bias
B3 Z-score & mean reversion
B4 AR(1) / Ornstein-Uhlenbeck & half-life
B5 Realised volatility
B6 Regime
B7 R^2 (goodness of fit)
B8 Correlation & collinearity
B9 PCA (factor structure)

C · Modelling concepts
C1 Fair value & the carry fit
C2 z_hist vs z_fair — the key distinction
C3 Features, target, train/test, walk-forward
C4 OU, Ridge, Gradient Boosting
C5 Overfitting, regularisation, bias-variance
C6 Hit rate, correlation, MAE

D · Trading concepts
D1 Signal vs trade candidate
D2 Direction: fade vs follow
D3 Transaction-cost model
D4 Expected net edge, regime multiplier, model agreement
D5 Signal score
D6 Position sizing, stop, max hold
D7 Harvest vs gated

E · Backtesting concepts
E1 Event backtest & walk-forward
E2 The three biases
E3 Every backtest metric

F · The pipeline, module by module

G · Results — is it profitable?

H · How to use it

I · Limitations & risks

J · One-line glossary

0 · What this is & the central question

This project reconstructs the *futures curve* of an asset, works out what that curve *should* look like from the economics of carrying the asset through time, measures where the real curve deviates from that, checks whether the deviation is a genuine opportunity or just noise, and backtests the resulting trades with realistic costs.

It deliberately follows a build order: **data → contracts → curve → basis → calendar spreads → curve shape → fair value → deviation → z-score → prediction → regime → cost/liquidity → risk → signal → backtest → live monitoring.**

> **The central research question is not** "are two futures prices close?" It is: *given maturity, carry, market regime, liquidity and historical behaviour, is the current relationship materially different from its estimated fair relationship — and is there enough expected net edge to justify the risk and execution cost?*

Three markets are wired today: **Gold** (COMEX dated futures), and **BTC** and **ETH** (Deribit perpetual + dated futures, where the funding rate is the carry analogue).

A · Market concepts

A1 · Spot & futures

Spot price — the price to buy or sell the actual asset *right now*, for immediate delivery. "Spot gold $2,500" = one ounce changes hands today for $2,500.

Futures contract — a standardised, exchange-traded agreement to buy or sell the asset at a *price locked in today* but with *settlement on a specified future date* (the expiry). Almost no cash moves today; you post *margin* (a good-faith deposit).

Futures price — the current market price of one such contract for a particular maturity. A futures price is **not** a forecast of the future spot price; it is today's spot adjusted for the cost and benefit of carrying the asset until expiry (see A6).

> **In this project:** Gold has no free live "cash" feed, so the nearest-dated contract is used as the spot anchor. Crypto *does* have a real spot index, so its basis is genuine.

A2 · Expiry & convergence

Expiry — the date a contract settles. COMEX gold terminates trading on the third-to-last business day of the delivery month; Deribit crypto quarterlies settle on the last Friday.

Convergence — as expiry approaches, a futures price is pulled toward the spot / settlement value, because there is less and less time left to "carry" the asset. At expiry the basis is zero.

> **In this project:** Convergence is why the nearest contract is the best spot proxy for gold, and why the engine force-closes any position when its near leg is within ~8 days of expiry.

A3 · Long, short, margin, mark-to-market

Long a future = you have agreed to *buy* at the locked price; you profit if the futures price rises. **Short** = you have agreed to *sell*; you profit if it falls. You can go short without owning the asset.

Margin — the deposit the exchange holds against your position (a few percent of contract value). **Mark-to-market** — each day the exchange revalues your position and moves cash between your account and the counterparty's, so gains and losses are realised daily.

> **In this project:** The backtests assume a $50,000 notional book and charge realistic margin-free P&L per 1-lot spread; leverage/margin limits are noted as a Phase-2 item.

A4 · Settlement: physical vs cash

Physically settled — at expiry the short delivers the actual asset (many commodity contracts). **Cash settled** — no asset moves; the contract pays the difference versus a settlement price. Index futures (NIFTY) and Deribit crypto futures are cash settled.

A5 · Basis

Basis = Futures price − Spot price. **Basis %** = (Futures − Spot) / Spot. Positive basis → futures above spot; negative → below.

```
basis      = F  −  S
basis_pct  = (F − S) / S
annualised = (F / S − 1) × 365 / days_to_expiry   ← comparable across maturities
```

> **In this project:** Gold: `basis = contract − nearest contract` (no cash feed). Crypto: `basis = dated future − index`, and separately `perp basis = perp − index` (reported in basis points).

A6 · Cost of carry (the r, s, y, T model)

Carry — the net cost of holding exposure to the asset until a future date. The no-arbitrage fair futures price is:

```
F(T) = S × (1 + (r + s − y) × T)          (simple form)
F(T) = S × exp((r + s − y) × T)          (continuous form the engine fits)
```

Symbol	Name	Meaning
`S`	spot	price of the asset now
`r`	financing rate	interest cost of the money to buy and hold the asset (per year)
`s`	storage	storage + insurance cost for a physical commodity (per year); ≈ 0 for crypto/index
`y`	convenience yield	benefit of holding the *physical* asset now — dividends for an index, scarcity premium for a commodity (per year)
`T`	time to expiry	in **years** (90 days → T = 0.2466)

Read it as: *futures price = spot, pushed up by what it costs to carry, pushed down by what you give up by not holding the real thing*. The single number `(r + s − y)` is the **net carry rate**; it sets the slope of the whole curve.

> **In this project:** `src/fairvalue.py` fits `ln F = a + c·T` across all contracts each day. The slope `c` is the **implied carry rate**; `exp(a)` is an implied spot. Gold's fit $R^2 \approx 0.999$ (the carry model explains gold's curve almost perfectly). For crypto the carry analogue is the funding rate (A13).

A7 · Contango & backwardation

Contango — later-dated futures priced *above* nearer ones (upward curve); the normal state when carrying costs dominate (gold, most of crypto in this sample). **Backwardation** — later-dated *below* nearer (downward curve);

happens when there is a strong benefit to holding the physical *now* (shortage, squeeze, high dividends).

A8 · Calendar spread

Calendar spread (a.k.a. time spread) — the price of one expiry minus another of the *same* underlying. Convention here: `later − earlier`, so positive = contango.

> **Why spreads are the heart of the engine.** A calendar spread is a paired position: long one expiry, short another. When the whole market jumps, both legs move together and the common move *cancels*. What is left is the *shape* of the curve — a much quieter, more mean-reverting quantity than an outright price. Gold's calendar spreads have std ≈ $4 while the price level swings by hundreds of dollars.

> **In this project:** Gold trades four adjacent spreads (`cs_Z_V` , `cs_G_Z` , `cs_J_G` , `cs_M_J`). Crypto trades `qbasis_<T> = dated future − perp` , the direct analogue.

A9 · Term structure / futures curve

Futures curve / term structure — the set of futures prices across all available expiries at one moment, plotted against time to expiry. Compare curves across dates by *days-to-expiry*, never by calendar month (a "March contract" means something different depending on today's date).

> **In this project:** `src/curve.py` cleans the raw data (drops zero-volume placeholder rows), joins every contract into a date×contract panel, and attaches each contract's `T` . The sorted (T, price) sequence for a date *is* that day's curve.

A10 · Level, slope, curvature

Factor	Meaning	Measure used
Level	the whole curve higher/lower	mean of the contract prices
Slope	steepness of contango/backwardation	`(F_far − F_near)/(T_far − T_near)` , $ per year
Curvature (butterfly)	the belly vs a straight line from near to far	`F(belly) − line_at_belly`

Almost every curve move is one of these three. `slope` is essentially the sum of the adjacent calendar spreads, so it is *not* independent information from them — feeding both to a model double-counts. Curvature *is* largely independent.

A11 · Roll yield / roll-down

Roll yield — the gain or loss from replacing an expiring contract with a later one. In contango a *long* pays the spread to roll (negative roll yield); a *short* collects it. As a contract ages, its price rolls down the curve toward spot ("roll-down").

> **In this project:** Shorting a crypto quarterly basis and holding it collects roll-down — but the harvest-vs-gated backtest (D7) shows roll-down *alone* loses money after fees; timing is what makes it pay.

A12 · Convenience yield

Convenience yield (`y` in A6) — the non-cash benefit of holding the physical asset now rather than a claim on it later: being able to use it, sell into a shortage, or meet a delivery obligation. High convenience yield pulls the curve toward backwardation. Gold's is ≈ 0, which is why its carry fit is so clean.

A13 · Perpetual swap & funding rate

Perpetual swap ("perp") — a crypto futures contract with *no expiry*. To keep its price tethered to spot, longs and shorts exchange a **funding rate** every 8 hours: when the perp trades above the index, longs pay shorts (positive funding), and vice versa.

Funding is the crypto **carry rate**. Annualised: `funding_ann ≈ funding_8h × 3 × 365`. A dated crypto future should sit above the index by roughly `index × funding_ann × T`.

> **In this project:** `fair_qbasis = index × funding_ann_7d × T` is the engine's fair value for how much a dated crypto future should exceed the perp. The gap between the real premium and this fair value is the tradeable signal.

A14 · Perp basis & quarterly basis

Perp basis = perp − index (small, in basis points; reflects instantaneous funding pressure). **Quarterly basis** = dated future − index, usually expressed annualised. **qbasis** in this project = dated future − perp — the premium of the quarterly over the perpetual, which is what you actually trade in a perp-vs-quarterly basis trade.

B · Statistics concepts

B1 · Rolling window & point-in-time

Rolling window — a statistic (mean, std) computed over the last *N* observations, recomputed each day. **Point-in-time (PIT)** — every number available on day *t* uses only data from day *t* or earlier. The engine lags rolling stats by one day: `series.rolling(N).mean().shift(1)`.

B2 · Look-ahead bias

Look-ahead bias — accidentally using information that would not have been known at the time (tomorrow's price, a full-sample mean, a model trained on the test period). It is the single most common way a backtest lies. Every rolling statistic here is shifted; every ML model is retrained walk-forward; fair value uses only that day's cross-section of prices.

B3 · Z-score & mean reversion

Z-score — how far a value sits from its recent mean, in standard-deviation units: `z = (x − rolling_mean) / rolling_std`. |z| ≈ 2 is a roughly 1-in-20 reading if the series were Gaussian and independent (it usually isn't).

Mean reversion — the tendency of a series to return toward its average after straying. A spread that mean-reverts can be faded; one that *trends* cannot.

> **A large z-score is not a trade — it is an investigation signal.** The engine's convergence study found gold's spreads had `p_revert < 0.5` in every z bucket over the sample: they trended, they did not revert. Fading them lost money. Measuring this instead of assuming mean reversion is the point.

B4 · AR(1) / Ornstein-Uhlenbeck & half-life

AR(1) — a series where today = ρ × yesterday + noise (after subtracting the mean). If 0 < ρ < 1 the series mean-reverts. The continuous version is the **Ornstein-Uhlenbeck (OU)** process.

Half-life = −ln(2) / ln(ρ) — the number of days to close half the gap to the mean. Gold's calendar-spread half-lives were 26–46 days (slow). A signal with a 35-day half-life and a 10-day holding period only expects to capture ~18% of its gap.

B5 · Realised volatility

Realised volatility — the standard deviation of recent daily returns, scaled to a yearly figure: `std(daily log returns, 20d) × √252` (×√365 for crypto). A measure of how turbulent the market is right now.

B6 · Regime

Regime — a label for the current market state: calm / normal / stressed (from a point-in-time percentile of realised vol), trending / range (from the 20-day return), and for crypto a funding regime (hot / neutral / negative). Relationships that hold in one regime break in another, so signals are conditioned on it.

B7 · R^2 (goodness of fit)

R^2 — the fraction of a series' variation explained by a model; 1.0 = perfect, 0 = no better than the mean. Gold's daily carry-curve fit has $R^2 \approx 0.9993$ — the cost-of-carry model essentially *is* the gold curve, which is why residual mispricings are tiny.

B8 · Correlation & collinearity

Correlation — how tightly two series move together (−1 to +1). **Collinearity** — when model inputs are near-duplicates of each other (e.g. slope and the sum of the spreads). It makes linear models unstable and their coefficients meaningless; the engine drops `slope_ann` from the ML features for exactly this reason.

B9 · PCA (factor structure)

Principal Component Analysis — finds the few underlying "directions" a set of correlated series moves along. Gold's four calendar spreads share one driver (the net carry rate), so a first component would explain ~90%+ of their variance and look like "all spreads widen/narrow together." Modelling 1–2 factors instead of 4–10 spreads reduces overfitting. (Implemented as a descriptive KMeans cluster today; full PCA is a Phase-2 item.)

C · Modelling concepts

C1 · Fair value & the carry fit

Fair value — a model estimate of what a contract or spread *should* be worth. Here: fit `ln F = a + c·T` across the whole curve each day (ordinary least squares). `c` = implied carry rate, `exp(a + c·T)` = fair price of each contract, `actual − fair` = the deviation. For crypto, fair value of the quarterly premium is `index × funding_ann_7d × T`.

C2 · z_hist vs z_fair — the key distinction

	`z_hist`	`z_fair`
Question	"far from its own recent average?"	"far from what the carry / funding model says it should be?"
Fails when…	the fair level genuinely moved (rates rose) — flags a false anomaly	the model is wrong or mis-specified
Verdict	weak — a rolling mean is a stale benchmark	the cleaner anomaly signal

Worked example (Gold, 2026-09-04). Every calendar spread showed `z_hist ≈ +1.3` ("spreads look wide"). But `z_carry = +1.6` — the carry rate itself had risen — and `z_fair ≈ ±0.4`. Conclusion: the steepness was a *real carry move, not a mispricing*. A naive z-score system would have flagged a trade and been wrong.

C3 · Features, target, train/test, walk-forward

Feature — an input the model sees (z-scores, curvature, carry rate, realised vol, days-to-expiry, 5-day momentum…). **Target** — what it predicts: here the *change in each calendar spread over the next 10 trading days*.

Walk-forward validation — train on an expanding window of the past, predict the next block, step forward, retrain. Never train on data that comes after the prediction date. This is the honest way to measure a time-series model; a random train/test split leaks the future.

C4 · OU, Ridge, Gradient Boosting

Model	What it is	Gold OOS result
OU	analytic mean-reversion: predict $-(1-\rho^{H})\times$(gap to mean)	no skill (hit 0.38) — spreads trended
Ridge	linear regression with an L2 penalty that shrinks coefficients toward zero (controls overfitting)	no skill (hit 0.44)
Gradient Boosting (GBR)	an ensemble of small decision trees, each correcting the last one's errors; captures non-linear interactions	**weak positive skill** (hit 0.54, corr 0.19)
Ensemble	average of models	washed out by the weak linear models

> **In this project:** Only GBR is used as a live tilt, and only as a weak prior — with ~240 days of data it is "a nudge, not a crystal ball."

C5 · Overfitting, regularisation, bias-variance

Overfitting — a model that memorises noise in the training data and fails out-of-sample. **Regularisation** — deliberately constraining a model (L2 penalty, shallow trees, few features) to prevent that. **Bias-variance trade-off** — too simple a model is biased (misses real structure); too complex a model has high variance (fits noise). Small samples (241–400 days here) force simple models.

C6 · Hit rate, correlation, MAE

Hit rate — fraction of predictions with the correct sign; 0.50 = coin flip. **Correlation** of predicted vs realised change — directional strength. **MAE** (mean absolute error) — average size of the miss, in the target's units.

D · Trading concepts

D1 · Signal vs trade candidate

Signal — evidence that a relationship is statistically or economically abnormal. **Trade candidate** — a signal that *also* survives cost, regime, liquidity, days-to-expiry and model-agreement filters, with positive expected net edge. **"The contracts are close" is never itself a signal.**

D2 · Direction: fade vs follow

Fade — bet the deviation reverts (short a rich spread, long a cheap one). **Follow** — bet the move continues (trade with a momentum forecast). The engine fades the fair-value gap by default; a separate backtest strategy (D) tries following the GBR forecast.

"Long the calendar spread" = buy the later-dated leg, sell the earlier one; profit if the spread widens. "Short the spread" = the reverse; profit if it narrows.

D3 · Transaction-cost model

Yahoo/Deribit give no live bid/ask, so cost is modelled from tick size and a liquidity tier:

```
one-way cost per contract = half the bid-ask spread + slippage + commission
gold  : bid-ask 1–4 ticks ($10/tick) by liquidity tier + 0.5-tick slippage + $2.50
crypto: ~4 basis points of notional per leg (Deribit-style taker fee)
a calendar-spread round trip touches 2 legs, in and out = 4 fills
```

> **In this project:** Gold round-trip ≈ 1.1 "gold points" per spread vs a spread std of ~4 points — costs eat most of any edge. Crypto round-trip ≈ $126 on an $80k notional — small relative to basis moves of 2–5%, but fatal if you overtrade (see D7).

D4 · Expected net edge, regime multiplier, model agreement

```
expected_gross = 0.5 × |fair-value gap|          (expect to close ~half the gap)
expected_net   = expected_gross − round-trip cost (− funding carry, for crypto)
regime_mult    = 1.0 calm/normal ; ×0.5 stressed ; ×0.6 strong trend
model_agree    = does the GBR forecast point the same way? (×1.0 yes / ×0.45 no)
adj_net        = expected_net × regime_mult × model_agree factor
```

D5 · Signal score (0–100)

A weighted blend used only for *ranking*: conviction (|z_fair|), model agreement, liquidity tier, days-to-expiry comfort, and the size of the adjusted net edge. A score of 87/100 is a decision-support ranking, **not** a probability of profit.

D6 · Position sizing, stop, max hold

One position per spread, at most 3 concurrent. Exit on: reversion (`|z| < 0.4`), maximum hold (25–30 days), an adverse stop, or the near leg approaching expiry. Sizing is 1 lot in the MVP; score-scaled sizing and portfolio risk limits are Phase-2.

D7 · Harvest vs gated (why the filter matters)

The cleanest proof the engine adds value. On BTC, two backtests over the same period, same exits, same costs:

Strategy	Trades	Win %	P&L / $50k	Profit factor	Sharpe	Max DD
Harvest — short *every* contango basis	187	29%	−17.7%	0.75	−0.80	−$15,633
Gated — only when `z_gap ≥ 1.25`	29	65%	+14.1%	1.68	0.84	−$7,115

Blindly harvesting the basis *loses* — you pay fees and funding constantly and get run over when the basis widens. Entering only when a dated future is *statistically rich versus funding* turns it positive. The signal, not the roll-down, is the edge.

E · Backtesting concepts

E1 · Event backtest & walk-forward

Event backtest — step through history day by day; each day, close positions whose exit conditions are met and open new ones from that day's candidates (ranked by score, capped at 3). Equity is marked daily (realised P&L + open mark-to-market). All decision inputs are point-in-time.

E2 · The three biases that kill backtests

- **Look-ahead bias** — using future information (see B2). Guarded by lagging every rolling stat and retraining walk-forward.
- **Survivorship bias** — testing only on instruments/contracts that still exist. Guarded by using each contract only over the dates it actually traded.
- **Data leakage** — the target or test period bleeding into training. Guarded by an expanding-window split with the target horizon respected.

E3 · Every backtest metric, defined

Metric	Meaning	Rule of thumb
Total P&L	sum of trade results, USD	positive, obviously — but see sample size
Return on $50k	total P&L ÷ assumed book	context for the P&L number
Win rate	fraction of trades that made money	> 0.5 helps but a low-win / big-winner profile can still work
Avg win / avg loss	mean P&L of winners vs losers	winners should not be dwarfed by losers
Profit factor	gross profit ÷ gross loss	> 1 = profitable; > 1.5 = decent; < 1 = loses
Sharpe (annualised)	mean daily P&L ÷ its std, ×√252 (×√365 crypto)	> 1 good, > 2 strong — *but meaningless on < ~30 trades*
Sortino	like Sharpe but only penalises downside volatility	> Sharpe usually
Max drawdown	largest peak-to-trough equity fall	can you stomach it? BTC gated: −$7,115
Avg hold days	mean time in a trade	should match the signal's half-life

Sample-size caveat. Gold: 241 trading days, 15–37 trades per strategy. Crypto: ~400 days, one funding regime, ~30 gated trades. These numbers are *directional evidence*, not validated estimates. A Sharpe of 0.84 on 29 trades has a wide confidence interval that includes zero.

F · The pipeline, module by module

File	Blueprint	Input → Output	Terms it produces
`src/fetch_data.py`	§7	Yahoo → `data/raw/*.csv` (immutable)	OHLCV per gold contract + front anchor
`src/contracts.py`	§8	symbol → object	underlying, expiry, days-to-expiry, T
`src/curve.py`	§9	raw → `curve_long.csv`	cleaned date×contract panel with T
`src/features.py`	§10	panel → `features.csv`	basis, calendar spreads, slope, curvature, level
`src/fairvalue.py`	§12	panel → `fairvalue.csv`	implied carry rate, fair curve, fair spreads, deviations, fit R^2
`src/deviation.py`	§11	features + fair → `deviation.csv`	`z_hist`, `z_fair`, `z_carry`, anomaly flags
`src/regime.py`	§17	panel → `regime.csv`	realised vol, calm/normal/stressed, trend, cluster
`src/convergence.py`	§27,§29	features + dev → `convergence.csv`	AR(1) half-life, empirical reversion table by z bucket
`src/predict.py`	§14,§16	all → `predictions.csv`	walk-forward OU / Ridge / GBR forecast of 10-day spread change
`src/signals.py`	§18,§19	all → `signals.csv`	fair gap, direction, cost, regime mult, adj-net edge, score, trade_candidate
`src/backtest.py`	§20	signals → `data/output/bt_*`	4 strategies, equity curves, all metrics in E3
`src/pipeline.py`	§6	—	runs the whole gold chain (~3 s)
`app/dashboard.py`	§22	reads outputs	Streamlit product view; sidebar market selector
`src/monitor.py`	§22	refresh → `alerts.json`	deviation / trade-candidate / curve-flip alerts

`src/crypto/fetch_crypto.py`	§5,§7	Deribit → `data/crypto/<CCY>/raw`	perp + dated OHLC, index, funding rate
`src/crypto/engine.py`	§9–20	raw → `data/crypto/<CCY>/{clean,output}`	`qbasis`, funding fair value, gap, `z_gap`, funding regime, signals, harvest-vs-gated backtest
`src/crypto_pipeline.py`	—	—	fetch + analyse BTC and ETH

G · Results — is it profitable?

Gold (COMEX) — 241 trading days

- Cost-of-carry model explains the curve to **$R^2 \approx 0.999$**; implied carry 4.0–4.7%/yr — economically sane.
- Convergence study: spreads **trended, did not revert** (`p_revert < 0.5` in every z bucket).
- Backtest, four strategies:

Strategy	Trades	Win %	P&L / $50k	Profit factor	Sharpe
A — naive `z_hist` fade	37	22%	−8.3%	0.19	−2.17
B — fair-value fade	24	33%	−3.6%	0.36	−2.18
C — gated (full discipline)	15	47%	−1.2%	0.60	−0.77
D — trade with the GBR forecast	63	32%	−8.3%	0.29	−1.54

Verdict: not profitable on gold, and the engine correctly says so ("stand aside"). Gold's curve is efficiently priced; residual mispricings (~2 pts) do not clear costs (~1.1 pts). The value delivered is rigorous measurement and capital protection: each discipline layer cuts the loss, and the naive strategy loses roughly **7× more** than the gated one.

BTC (Deribit) — ~400 days, one funding regime

Strategy	Trades	Win %	P&L / $50k	Profit factor	Sharpe	Max DD
Harvest — every contango basis	187	29%	−17.7%	0.75	−0.80	−$15,633
Gated — `z_gap ≥ 1.25`	29	65%	**+14.1%**	1.68	0.84	−$7,115

Verdict: a real but modest edge, conditional on selectivity. The gated strategy is profitable in-sample with profit factor 1.68; the harvest baseline proves the z-score filter — not roll-down — is what makes it work. Caveats: Sharpe < 1, a −$7k drawdown on a $50k book, only 29 trades, and a single year that was mostly one funding regime. This needs out-of-sample and paper-trade validation before real capital.

ETH (Deribit) — ~400 days

Strategy	Trades	Win %	P&L / $50k	Profit factor	Sharpe
Harvest	165	35%	−0.1%	0.97	−0.09
Gated	33	64%	+0.7%	2.26	1.15

Verdict: essentially flat. ETH's curve sits very close to funding-fair value (carry-vs-funding gap only ~0.5%), so even the gated strategy captures almost nothing after fees. The engine correctly issues few or no candidates. This is a feature: it does not manufacture trades where none exist.

What "profitable" means here, and the conditions

- All figures are **in-sample, single-period, small-N**. "Profitable" here means "showed positive net P&L after modelled costs on the available history" — a hypothesis, not a track record.
- The BTC edge would likely **persist** if: funding-vs-basis dislocations keep recurring (they have historically), fees stay near 4 bps/leg, and the strategy stays selective (~2–4 trades/month).

- It would likely **decay** if: crypto term structure becomes as efficient as gold's, funding regimes shift and stay shifted, or position size grows enough to move the thinner dated contracts.
- Gold is a standing example of an efficiently-priced curve where the correct output is *no trade*. That the engine reaches that conclusion — rather than curve-fitting a fake winner — is the main evidence it is built honestly.

H · How to use it

```
pip install -r requirements.txt

python demo.py                     # runs gold + crypto, prints the results table
python src/pipeline.py             # gold only: fetch → … → backtest  (~3 s cached)
python src/crypto_pipeline.py      # BTC + ETH via Deribit (no API key)
python -m streamlit run app/dashboard.py   # product view — sidebar: Gold / BTC / ETH
python src/monitor.py --no-refresh # gold alert digest → data/output/alerts.json

# or:  make setup | make demo | make gold | make crypto | make dashboard | make test
```

Dashboard tabs. Gold: Curve & Fair Value · Spreads & Deviation · Regime · Signals · Backtest · Convergence. Crypto: Curve & Funding · Basis vs Funding · Signals · Backtest (harvest vs gated).

Reading a verdict. "STAND ASIDE" = no relationship is abnormal enough, or expected edge does not clear costs. "TRADE CANDIDATE" = a ranked list with direction, score, and expected net $/lot — evidence to investigate, not an instruction.

Extending to a new market (Silver, NIFTY, …): add a fetch adapter that writes the same `data/raw` CSV schema; add the contract's month-code and expiry rule to `src/contracts.py`; for anything with a funding rate, add it as a feature and a carry term. Everything downstream is unchanged.

I · Limitations & risks

- **Data.** Free sources only. Gold has no true cash feed (nearest contract used as anchor); no open interest from Yahoo. Crypto history is ~1 year and one funding regime.
- **Sample size.** 15–40 trades per backtest. Statistical error bars are large; a positive Sharpe here is not significant at any conventional level.
- **Costs.** Modelled from tick size / fee schedules, not live bid-ask. Real slippage in thin dated contracts could be worse.
- **No live validation.** Nothing here has been paper-traded or run forward.
- **Model risk.** The carry and funding fair-value models are simplifications; a regime they were not fit on can break them.
- **Execution.** Margin, leverage limits, funding settlement timing, and exchange outages are not modelled in the MVP backtest.

Disclaimer. This is a research and decision-support system. It is not investment advice and makes no promise of profit. Validate every result out-of-sample and paper-trade before risking capital. Trading futures involves substantial risk of loss.

J · One-line glossary

Spot	price to transact the asset now
Future	agreement to transact at a locked price on a future date
Expiry	the date a contract settles
Convergence	futures → spot as expiry nears; basis → 0
Basis	future – spot
Carry / cost of carry	net cost of holding the asset through time = financing + storage – convenience yield
Convenience yield	benefit of holding the physical asset now
Contango / backwardation	curve sloping up / down
Calendar spread	price of one expiry minus another, same underlying
Term structure / curve	futures prices across all maturities at one moment
Level / slope / curvature	the three ways a curve moves
Roll yield / roll-down	gain/loss from replacing an expiring contract; price sliding down the curve
Perpetual swap	crypto future with no expiry, tethered by funding
Funding rate	periodic payment between perp longs and shorts; the crypto carry rate
qbasis	dated crypto future – perpetual
Fair value	model estimate of what a price/spread should be
Implied carry rate	slope of the fitted `ln F` vs `T` line
Z-score	distance from mean in standard-deviation units
z_hist / z_fair	deviation vs recent history / vs the carry-fair model
Mean reversion	tendency to return toward the average
AR(1) / OU	simplest mean-reverting process
Half-life	days to close half the gap to the mean
Realised volatility	annualised std of recent daily returns
Regime	label for current market state (calm/stressed, trend/range, funding hot/neutral)
Point-in-time	using only information available at the time
Look-ahead bias	accidentally using future information
Walk-forward	train on the past, test on the next block, repeat
Ridge	linear regression with a coefficient-shrinking penalty
Gradient boosting	ensemble of small trees, each fixing the last's errors
Overfitting	memorising noise; fails out-of-sample
Hit rate	fraction of directionally-correct predictions
Signal vs trade candidate	abnormality vs abnormality that survives cost/regime/risk filters

Fade / follow	bet on reversion / bet on continuation
Signal score	0–100 ranking blend; not a probability of profit
Profit factor	gross profit ÷ gross loss
Sharpe / Sortino	return per unit of total / downside volatility
Max drawdown	largest peak-to-trough equity fall
Harvest vs gated	trade every basis vs only statistically-rich ones

Generated for the Futures Curve Intelligence Engine project. Figures are from the in-sample backtests described above and will change as data updates. Research use only.